AKASH K PILLAI

Backend Developer | AI Developer

+91-73563-44191 | akashkpillai55@gmail.com | Alappuzha, Kerala, India
linkedin.com/in/akash-k-pillai | github.com/akashkpillai

PROFESSIONAL SUMMARY

Backend and AI Developer with 3+ years of experience building REST APIs and backend applications. Skilled in Node.js, NestJS, PostgreSQL, and MongoDB, with hands-on experience in LLM integration, RAG, vector search, and tool calling. Experienced in leading backend delivery and building reliable, access-controlled AI workflows.

PROFESSIONAL EXPERIENCE

IITIL **Remote**
Innovative Information Technologies India Limited
Software Developer **July 2026 - Present**

- Led end-to-end backend development for an HRMS application using NestJS and PostgreSQL, implementing REST APIs and core business logic.
- Implemented notification services and Nodemailer email delivery; created Swagger API documentation for frontend integration.
- Conducted project demonstrations, resolved frontend integration issues, and guided new developers through the codebase and API workflows.
- Built an OpenAI-powered HRMS assistant for employee information, date-specific leave enquiries, and upcoming holidays.

INFINEIT GLOBAL SERVICES LLP **Surat, Gujarat, India**
Backend Developer **March 2023 - July 2026**

- Built and maintained RESTful APIs and backend services using Node.js, Express.js, NestJS, and Sails.js for production web applications.
- Led backend development using NestJS, Prisma, and PostgreSQL, including schema design, database integration, and API architecture.
- Optimized API performance and database queries; tested services with Postman and Jest.js and resolved critical production bugs.
- Mentored 2 junior developers on backend best practices, code quality, and Git workflows.

TECHNICAL SKILLS

Languages & Backend: JavaScript, TypeScript, Node.js, NestJS, Express.js, Sails.js, REST APIs

AI & Retrieval: OpenAI API, Groq API, RAG, pgvector, embeddings, vector search, tool/function calling, conversational assistants, custom agent workflows

Databases & Architecture: PostgreSQL, MongoDB, Prisma, Mongoose, Redis, Kafka, microservices, event-driven architecture

Quality & Security: Jest, API testing, JWT, bcrypt, RBAC, deterministic evaluations, citation validation, bounded retries, agent step limits, token/cost tracking

Tools & Integrations: Git, GitHub, Docker, Docker Compose, Swagger/OpenAPI, Postman, Jira, Nodemailer, EJS, Razorpay, PayPal, Stripe, Twilio, TMDB, YouTube API

Frontend: React.js, HTML, CSS, Tailwind CSS, Material UI

PROJECTS

Meridian Assistant - Discord-Based AI Assistant

Independent Project 2026
Technologies: TypeScript, Node.js, OpenAI API, PostgreSQL, pgvector, Discord.js, Docker

- Built a Discord assistant with RAG, vector search, source citations, and a custom tool-calling agent for document search and proposal estimates.
- Enforced brand-scoped access in application code and SQL queries; used private Discord replies to protect confidential information.
- Implemented immutable estimate versions and DOCX generation gated by explicit approval of the exact stored version.
- Added step limits, bounded retries, and token/cost logging; created 30 deterministic evaluation cases for retrieval, authorization, prompt-injection scenarios, approvals, and failures.

HRMS & AI Assistant

IITIL July 2026 - Present
Technologies: NestJS, TypeScript, PostgreSQL, OpenAI API, Nodemailer, Swagger/OpenAPI

- Built REST APIs, core HR workflows, notifications, and email delivery; documented APIs and supported frontend integration and developer onboarding.
- Developed an OpenAI-powered assistant to retrieve employee details, check leave on specific dates, and list upcoming holidays.

FineOne - Unified Diamond Platform

InFineIT Global Services LLP 2025 - Present
Technologies: NestJS, PostgreSQL, Prisma, Redis, Kafka, Microservices, Docker

- Developed backend modules and REST APIs; implemented Redis caching across microservices and designed a notification service.
- Designed relational schemas and raw PostgreSQL queries, conducted code reviews and unit testing, and led internal technical learning sessions.

FineStar - Diamond Stock Management Platform

InFineIT Global Services LLP 2023 - 2025
Technologies: Sails.js, Node.js, MongoDB, Mongoose, Nodemailer

- Built role-based APIs for diamond stock filtering and user analytics; implemented a bidding system with validation and scheduled bidding windows.
- Created scheduled email notifications with reusable EJS templates and optimized MongoDB aggregation queries for reporting.

AI-Powered Resume Analyzer

Personal / Internal Tool github.com/Akashkpillai/resume-analyzer
Technologies: NestJS, Groq API

- Delivered an AI-powered resume analyzer to the HR team to support the internal candidate screening workflow.

EDUCATION

Bachelor of Science (B.Sc.) in Physics

NSS College, Kerala University | Pandalam, Pathanamthitta, Kerala, India | 2018 - 2021

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Intermediate), Malayalam (Native)